Amogh Paruvelli

Software Engineer
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Education

University of Wisconsin-Madison Madison, Wisconsin
Bachelor of Science in Computer Science and Data Science *September 2023 – December 2026*
GPA: 3.58

Relevant Coursework: Relational Databases, Statistical Modeling, Artificial Intelligence, Operating Systems, Matrix Methods in ML, Deep Learning for NLP, Multivariate Calculus, Linear Algebra, Data Structures, Algorithms, Capstone in Computer Science

Skills

Full-Stack: Python, FastAPI, TypeScript, JavaScript, React, Node, Java, Spring MVC

AI & LLM: PyTorch, RAG, MCP (Model Context Protocol), Pinecone, Pydantic

Data & Analytics: SQL, PostgreSQL, Redis, Databricks, PySpark

Cloud & Tools: AWS (Lambda, API Gateway, S3, Bedrock, Amplify), Docker, CI/CD, Git, Codex, Claude Code

Experience

Data Engineer Intern May 2026 – August 2026
Ebara Elliott Energy *Jeannette, Pennsylvania*

- Slashed data ingestion time 95% (60 min to 3 min) by building a self-service ETL web-app with Python/Streamlit and Apache Spark that loads CSV, JSON, Parquet files into Databricks tables, automating 4-5 person manual ticket process
- Designed master schema with 2,082+ golden records from 15 sources, consolidating 8.2k record linkage conflicts by implementing RapidFuzz pipeline with TF-IDF weighted fuzzy matching and calibrated log-odds thresholds

Software Engineer February 2026 – July 2026
Handlab Poker Intelligence *Remote*

- Built RAG engine for 30k transcribed poker games, expanding natural language queries against 400-term jargon dictionary and retrieving relevant strategic trends across hand histories indexed via Pinecone
- Engineered custom SQL translation layer with 3s response time for exact betting line searches, assessing user intent from prompts via LLM, then generating JSON request bodies for API to dynamically construct queries and execute
- Shipped AI coaching SaaS tool to **3 paying clients**, iterating on feedback via monthly Agile sprints in two-person team

Software Engineer Intern June 2025 – September 2025
Spencer's Gifts and Spirit Halloween *Egg Harbor Township, New Jersey*

- Implemented Java endpoint class returning per-variant stock status and pickup button title and message text as JSON for the shopper's selected store, serving 250k daily users ahead of peak Halloween
- Supported site reliability under 4x traffic load by cutting redundant inventory API calls from product pages, writing the Redis key builder and GET and SET calls for a cache-aside lookup
- Wrote **12 JUnit tests** on a mocked Redis client covering controller validation and service caching, plus cache-miss counters
- Gained working proficiency in enterprise Java by building employee and department CRUD service with Spring MVC and JDBC on MS SQL Server, then refactoring into controller, service and repository layers with Hibernate ORM

Projects

Find the Outfit (Amazon Shopbop Capstone) | *TypeScript, Expo, AWS Lambda, Amazon Bedrock* September 2026 – Present

- Launched serverless backend backed by **116 Vitest unit tests**, serving GET /puzzle/today and POST /guess on AWS Lambda and API Gateway, rating each guess warmer or colder by cosine similarity of Bedrock Titan embeddings
- Ran weekly Agile sprints and Jira ticketing as Scrum Master for a 5-person team building a Wordle-style puzzle where shoppers guess a hidden Shopbop item from a cropped photo in six tries, live on AWS Amplify

SAW (Shared Agent Workspace) | *TypeScript, Node.js, MCP, Claude Code, SQLite, Docker* July 2026 – Present

- Exposed chat messages and git refs through **5 MCP tools** by building multi-agent coordination layer that lets AI coding agents (Codex, Claude Code) pair-program in a shared workspace over a custom WebSocket relay protocol
- Containerized relay service with Docker, persisting SQLite state to a mounted volume across restarts, running self-hosted

Curio | *FastAPI, OpenAI API, WeasyPrint, React, Tailwind CSS, Vercel* October 2024 – January 2025

- Generated computer science curricula in under 60 seconds with full-stack platform built on FastAPI and React, integrating GPT-4o endpoints to automate localized problem-set creation; shipped MVP at MadHacks 2024
- Debugged 13 null-pointer worksheet errors caused by unstructured model outputs, using Pydantic to enforce response validation across 5 defined question-pattern JSON schemas before results reached frontend

Research

Disentanglement of Safety and Utility in LLM Activation Spaces | *PyTorch* January 2026 – May 2026

- Bypassed AI guardrails with a 93% attack success rate across 420 harmful prompts via activation-vector ablations, retaining 62% reasoning accuracy by extracting utility direction from 37k HelpSteer response pairs
- Led team of 5 to reverse-engineer how Gemma-2-2B-IT and Llama-3.1-8B-Instruct balance safety rules against helpful responses, replicating Difference-of-Means and ActSVD methods from two arXiv publications